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Characterizing Pediatric Myalgic Encephalomyelitis/Chronic Fatigue Syndrome: Symptom Burden, Functional Impairment, and Comparison With Adults in the Multi-Site Clinical Assessment of ME/CFS (MCAM) Study

Stephanie Sargent Weaver, PhD, MPH, Joanna J. Regan, MD, MPH, Anindita Issa, MD, Elizabeth A. Fall, MPH, Yang Chen, DrPH, Jin-Mann S. Lin, PhD, Elizabeth R. Unger, MD, PhD

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Characterizing Pediatric Myalgic Encephalomyelitis/Chronic Fatigue Syndrome: Symptom Burden, Functional Impairment, and Comparison With Adults in the Multi-Site Clinical Assessment of ME/CFS (MCAM) Study

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Division of High-Consequence Pathogens and Pathology, Centers for Disease Control and Prevention, Atlanta, GA 30329, USA

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Corresponding Author:

Stephanie Sargent Weaver, PhD, MPH

1600 Clifton Rd NE

Roybal Building 24

Mail Stop H24-12

Centers for Disease Control and Prevention

Atlanta, GA 30333

E-mail: SWeaver2@cdc.gov

Phone: 770-375-6961

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Declaration of Competing Interest

The authors declare no competing interests.

Disclaimer

The findings and conclusions in this report are those of the authors and do not necessarily represent the official position of the Centers for Disease Control and Prevention (CDC).

Ethics statement

The MCAM study was approved by the institutional review boards of the CDC, Open Medicine Institute Consortium, Mount Sinai Beth Israel, and Institute for Neuro Immune Medicine. Adult participants provided informed consent. For pediatric participants, written parental permission and participant assent were obtained.

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Objectives To characterize the clinical, functional, and educational characteristics of adolescents with myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) and compare selected illness domains with adults in the Multi-Site Clinical Assessment of ME/CFS (MCAM) study.

Study design Baseline cross-sectional data from adolescents aged 10–17 years enrolled in MCAM were analyzed. Standardized measures included the Short Form-36 (SF-36v2), CDC Symptom Inventory (CDC-SI), Multidimensional Fatigue Inventory (MFI-20), and selected domains of the DePaul Questionnaire. Descriptive statistics characterized the pediatric cohort and bivariate analyses compared adolescents with adults enrolled in MCAM.

Results Forty adolescents (mean age 15.3 years; 52.5% female) were included. Mean illness duration was 4.0 years. Educational disruption was substantial: 64% received homebound or homeschool instruction, 89% missed school because of illness, and participants missed a mean of 10.8 school days during the preceding month. Physical health-related quality of life was substantially impaired. Persistent post-exertional malaise (PEM), unrefreshing sleep, fatigue, and neurocognitive symptoms were common. Compared with adults (n = 595), adolescents had shorter illness duration and better physical functioning, vitality, and social functioning, whereas mental health scores were similar. Adults reported greater burden across several physical symptom domains although many between-group differences were small.

Conclusions Adolescents with ME/CFS experience substantial multisystem symptom burden, impaired physical functioning and educational disruption. Many core clinical features were similar to those in adults, supporting the need for early recognition and appropriate clinical and educational support.

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Keywords: Myalgic encephalomyelitis, Chronic fatigue syndrome, Adolescents, Post-exertional malaise, School absenteeism, Health-related quality of life, Neurocognitive symptoms

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Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a chronic multisystem illness characterized by substantial functional impairment associated with persistent fatigue, post-exertional malaise (PEM), unrefreshing sleep, cognitive impairment, and orthostatic intolerance [1,2]. Although most research has focused on adults, ME/CFS also affects children and adolescents and can result in significant disability during critical developmental years. Prior studies suggest illness onset in adolescents commonly follows an acute infectious illness, particularly infectious mononucleosis, although some patients experience a gradual onset without an identifiable trigger [3,4]. Prevalence estimates suggest that ME/CFS affects approximately 0.1%–0.4% of adolescents [5].

Previous studies have shown that adolescents with ME/CFS experience many of the same core clinical features observed in adults, including persistent fatigue, post-exertional malaise (PEM), unrefreshing sleep, cognitive dysfunction, headaches, musculoskeletal pain, and autonomic symptoms such as orthostatic intolerance and dizziness [1–3,9]. The illness is associated with marked reductions in health-related quality of life, prolonged school absenteeism, educational disruption, and reduced participation in social and recreational activities [6–8]. Despite a growing body of pediatric ME/CFS research, most studies have been conducted at single centers, and comprehensive multisite characterization using standardized measures remains limited, particularly among adolescents evaluated in specialty clinics [6–10].

Clinical recognition of pediatric ME/CFS can be challenging because symptom presentation spans multiple organ systems, diagnosis relies on self-reported symptoms using established diagnostic criteria, and alternative medical and psychiatric conditions must first be excluded [1,3,9]. In adolescents, diagnostic evaluation may be further complicated by normal developmental changes, including puberty, evolving sleep patterns, and increasing academic and

social demands [2]. Given these challenges, standardized characterization of adolescents with ME/CFS is essential to better define the illness and facilitate comparisons across patient populations.

The Centers for Disease Control and Prevention (CDC) initiated the Multi-site Clinical Assessment of Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (MCAM) study to characterize patients receiving care at specialty clinics with expertise in ME/CFS using standardized clinical instruments [10]. The pediatric MCAM cohort provides a unique opportunity to characterize the multidomain illness burden among adolescents evaluated in specialty settings. The primary objective of this study was to describe the clinical, functional, and educational characteristics of adolescents with ME/CFS enrolled in MCAM at baseline. A secondary objective was to compare selected illness domains between adolescents and adults evaluated at specialty clinics. Because participants had already been evaluated and were receiving care in ME/CFS specialty practices, the clinical features described in this study represent prevalent symptoms and functional status at the time of study assessment rather than presenting symptoms at illness onset or initial clinical evaluation. By providing standardized multisite data, this study aims to improve understanding of pediatric ME/CFS and inform clinical recognition and management.

Methods

Study Overview

CDC launched the Multi-site Clinical Assessment of Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (MCAM; CDC IRB# 6286) in 2012 to systematically characterize adults with ME/CFS evaluated in specialty clinical practices [10]. The systematic MCAM approach was adapted to collect longitudinal data on pediatric patients (10-17 years of age) with ME/CFS

(CDC IRB#6528) using a rolling cohort design over a 5-year period (2013-2020). This report uses the baseline data from both pediatric and adult MCAM studies. The MCAM study was approved by the institutional review boards of the CDC, Open Medicine Institute Consortium, Mount Sinai Beth Israel, and Institute for Neuro Immune Medicine.

Study Sites

Four clinics with specialized expertise in ME/CFS recruited pediatric participants: Institute for Neuro Immune Medicine, Open Medicine Clinic, Bateman Horne Center, and Weill Cornell Medicine. Adult participants were recruited from seven MCAM specialty clinic sites described previously [10].

Enrollment Criteria

Pediatric Cohort

Adolescents aged 10–17 years who were managed as having ME, CFS, or post-infective fatigue (PIF) at participating specialty clinics were eligible for enrollment. Exclusion criteria included human immunodeficiency virus (HIV) infection, or current pregnancy that precluded participation or informed consent. Eligible participants were contacted before scheduled clinic visits and could complete selected questionnaires in advance. Telephone consent was followed by written parental permission and participant assent at the clinic or through the study's web-based consent system.

Adult Cohort

Enrollment criteria and procedures for the adult MCAM cohort have been described previously [10]. Briefly, adults aged 18–70 years who were managed as having ME, CFS, or post-infective fatigue at participating specialty clinics were eligible for enrollment. Exclusion criteria included illness onset after age 62 years, HIV infection, current pregnancy, or dementia.

Sample Size

The final pediatric sample size was 40 with 19 males and 21 females. The adult sample size was 595 with 162 males and 433 females.

Data Collection

Data collection procedures were similar for both cohorts, although several questionnaires were adapted for pediatric participants. Participants completed standardized questionnaires assessing demographic characteristics, clinical history, functional status, fatigue, symptom burden, and health-related quality of life. Race and ethnicity were reported by the participant or parent/legal guardian, as applicable, using study questionnaire categories and were collected as separate variables to describe the study sample and facilitate comparison between the pediatric and adult cohorts. Selected clinical characteristics and medical history were obtained at enrollment. Table I summarizes the instruments used to assess the clinical, functional, and educational domains evaluated in this study. Unless otherwise specified, symptom and functional measures reflect participants' status at the time of the baseline MCAM assessment.

Measures

CDC Symptom Inventory (CDC-SI)

The CDC-SI assesses the occurrence, frequency, intensity, and duration of symptoms associated with ME/CFS [11]. The pediatric version emphasized case-defining symptoms and stomach or abdominal pain and included a 3-month duration option to permit comparison of pediatric case definitions. Higher scores indicate greater symptom burden [12].

Short Form-36 Health Survey (SF-36v2)

The Medical Outcomes Study Short Form-36 version 2 (SF-36v2) assesses health-related quality of life across eight domains: physical functioning, role limitations due to physical health, bodily

pain, general health perceptions, vitality (energy/fatigue), social functioning, role limitations due to emotional problems, and mental health [13,14]. Raw domain scores were transformed to norm-based T-scores (mean = 50, SD = 10 for the general population) according to the SF-36v2 scoring manual. Higher T-scores indicate better health status and functioning, and norm-based T-scores were used in all analyses.

Multidimensional Fatigue Inventory (MFI-20)

The MFI-20 is a 20-item self-report instrument that measures five dimensions of fatigue: general fatigue, physical fatigue, mental fatigue, reduced motivation, and reduced activity [15,16]. Subscale scores range from 4 to 20, with higher scores indicating greater fatigue.

DePaul Questionnaire

Neurocognitive and autonomic symptoms were assessed using selected items from the DePaul Pediatric Health Questionnaire (DPHQ) in adolescents and the DePaul Symptom Questionnaire in adults [17-20]. Composite symptom severity scores ranged from 0 to 16, with higher scores indicating greater symptom burden.

Statistical Analysis

The analytic sample included all adolescents enrolled in the MCAM study who met the eligibility criteria and had baseline data available for analysis. Because this was a secondary analysis of an existing cohort, no formal sample size calculation was performed.

Analyses were conducted using SAS version 9.4 (SAS Institute Inc., Cary, NC). Continuous variables are presented as means and standard deviations (SD), and categorical variables as frequencies and percentages. Percentages were calculated using available responses for each variable.

For analyses of symptom presence (prevalence within the study cohort), symptom variables were harmonized across instruments to represent symptoms present for at least 6 months. For CDC-SI variables, symptoms were considered present if participants reported experiencing the symptom during the preceding month and indicated that it had persisted for at least 6 months. Harmonized 6-month symptom variables available in the MCAM dataset were used for DPHQ and DSQ analyses.

Bivariate comparisons between adolescents and adults were conducted for selected illness domains assessed by the CDC-SI, SF-36v2, MFI-20, and DPHQ/DSQ. Categorical variables were compared using chi-square tests, and continuous variables were compared using Student's t tests or Mann-Whitney U tests, as appropriate. Statistical significance was defined as $P < 0.05$.

To aid interpretation of the magnitude of differences between groups, effect sizes for continuous outcomes were calculated using Hedges' *g*, which adjusts for unequal group sizes and small sample bias [21]. Effect sizes of approximately 0.20, 0.50, and 0.80 were interpreted as small, moderate, and large, respectively [22], and were considered alongside *p* values when interpreting group differences.

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Results

Participant Characteristics

Forty adolescents aged 10–17 years with ME/CFS were enrolled (mean age 15.4 years [SD 1.9]); 52.5% were female (Table II). Most participants identified as White (81.0%) and non-Hispanic (96.9%). The mean grade level was 9.5 (SD 2.0). All responding parents or legal guardians reported health insurance coverage and no unemployment benefits, and most primary guardians were married (81.1%) and employed full-time (72.7%).

The adult comparison cohort included 595 participants with ME/CFS (mean age, 47.5 years [SD 12.9]), of whom 72.8% were female. Most adults identified as White (88.2%) and non-Hispanic (79.3%), 93.8% had health insurance coverage, and 69.7% were not working at the time of enrollment.

Illness Characteristics

Among adolescents, the mean age at fatigue onset was 11.7 years (SD 2.5), and the mean age at ME/CFS diagnosis was 13.7 years (SD 2.8), representing an average interval of approximately two years between illness onset and diagnosis (Table III). Age at ME/CFS diagnosis and age at fatigue onset were available for 24 and 26 adolescents, respectively; these variables were missing or not reported for the remaining participants. Sudden illness onset was reported by 55.6% of adolescents and 63.2% of adults; this difference was not statistically significant (P = .358). Mean BMI was lower among adolescents than adults (21.9 [SD 4.6] vs 26.6 [SD 6.1]). Adolescents attended a mean of 2.1 specialty clinic visits during the preceding year, compared with 4.7 clinic visits among adults.

Functional Impact and School Participation

Functional impairment among adolescents was substantial (Table III). Nearly two-thirds (64%) received homebound or homeschool instruction, 89% missed school because of illness, and participants missed a mean of 10.8 school days over the past month (SD 8.3). Illness limited educational (85%), social (80%), and recreational (92.3%) activities for most participants. Rest provided little or no improvement in fatigue for 62.5% of adolescents.

Adults likewise reported substantial functional limitations. Illness limited social (96.4%) and recreational (97.6%) activities for most adults, and 74.6% reported limitations in work,

educational, or daily activities. Similarly, 60.0% reported little or no improvement in fatigue with rest.

Functional Status and Well-Being

Adolescents demonstrated substantial impairment in physical health-related quality of life but generally reported better physical health than adults with ME/CFS (Table IV). Moderate differences favored adolescents for vitality (g = 0.64), physical component summary (g = 0.50), and social functioning (g = 0.48). Smaller differences were observed for physical functioning (g = 0.42), role physical (g = 0.35), and bodily pain (g = 0.30). Mental health domains showed negligible differences between groups ($|g| < 0.15$).

Fatigue and Post-Exertional Malaise (PEM)

Persistent post-exertional malaise (PEM) was highly prevalent in both groups but was more commonly reported among adults than adolescents (87.7% vs 66.7%, P = 0.0002) (Table V). Adults also reported greater PEM severity, with a moderate effect size (g = -0.64). As expected, illness duration differed between groups, with adults reporting a much longer duration of illness than adolescents (13.6 vs 4.0 years; g = −0.96), representing the largest effect observed in the study. On the Multidimensional Fatigue Inventory (MFI-20), adults reported greater general fatigue (g = -0.65), physical fatigue (g = -0.51), and reduced activity (g = -0.45), whereas mental fatigue and reduced motivation were similar between groups (Table V). When asked to identify their most bothersome symptom, fatigue/tiredness/exhaustion was selected by 43.6% of adolescents and 61.9% of adults.

Neurocognitive and Autonomic Symptoms

Persistent neurocognitive symptoms were common among adolescents, particularly difficulty thinking or concentrating (67.5%) and forgetfulness or memory problems (55.0%)

(Table VI). Adults more frequently reported several neurocognitive symptoms, although these two symptoms did not differ significantly between groups.

Although adults generally reported a broader range of persistent symptoms, differences in symptom severity were mostly small. The largest difference was observed for dizziness (g = 0.69), with adolescents reporting greater symptom severity. Small-to-moderate differences favoring adolescents were also observed for difficulty paying attention (g = 0.42), irregular heartbeats (g = 0.40), and problems remembering things (g = 0.36). Conversely, adults reported somewhat greater severity for difficulty finding words (g = −0.33) and difficulty focusing on one thing (g = −0.22). Most remaining differences in symptom severity were small or negligible.

Pain, Immunologic, and Other Symptoms

Persistent muscle pain (57.5%), joint pain (51.3%), and headaches (52.5%) were common among adolescents (Table VII). Adults were more likely to report persistent muscle pain than adolescents (79.9% vs 57.5%, P = 0.0009) and had greater muscle pain severity (g = -0.57), whereas persistent joint pain and headaches did not differ significantly between groups (Table VII).

Adolescents frequently reported sensitivity to smells, foods, or chemicals (72.5%), persistent stomach or abdominal pain (47.5%), and upset stomach (60.0%) (Table VIII). Adults and adolescents did not differ significantly in the prevalence or severity of gastrointestinal symptoms.

Persistent unrefreshing sleep was reported by 72.5% of adolescents and 87.0% of adults (P = 0.0103). Adults also reported modestly greater symptom severity than adolescents (g = -0.39).

Discussion

The findings of this study support previous observations that adolescents with ME/CFS exhibit many of the same core clinical features as adults, including post-exertional malaise (PEM), unrefreshing sleep, fatigue, cognitive symptoms, and substantial functional impairment. Although adults generally reported longer illness duration and greater symptom burden across several physical domains, many between-group differences were small in magnitude, suggesting substantial similarity in the core clinical phenotype across age groups.

Consistent with previous studies reporting that pediatric ME/CFS often begins following an acute infectious illness, sudden illness onset was more common than gradual onset in our cohort, although a substantial proportion reported gradual onset. Adolescents also demonstrated substantial educational disruption, high rates of persistent PEM and unrefreshing sleep, marked fatigue, and reduced physical functioning and vitality [2,6–9]. The nearly equal sex distribution observed in the pediatric cohort contrasts with the marked female predominance among adults and is consistent with previous pediatric studies [2,9]. As expected, adolescents also had shorter illness duration, which may partially explain some observed age-group differences.

PEM was highly prevalent in both cohorts but was less frequent and less severe among adolescents. Adults also reported greater general and physical fatigue and lower vitality. Because the study was cross-sectional, these differences cannot be interpreted as evidence of disease progression. Nevertheless, the high prevalence of persistent PEM, unrefreshing sleep, and fatigue among adolescents demonstrates that key disease-defining features are present relatively early in the course of illness. Most adolescents reported little or no improvement in fatigue with rest [2,9].

Adolescents experienced substantial neurocognitive, autonomic, and pain-related symptom burden, although adults more frequently reported several symptoms. Differences in

symptom severity were generally modest, and some symptoms, including dizziness, were more severe among adolescents. These findings suggest that symptom prevalence and severity provide complementary information and that the relative prominence of symptom domains may vary with age. Cognitive dysfunction, autonomic symptoms, and pain have been identified previously as important contributors to functional impairment in pediatric ME/CFS [2,6,8,9].

Educational disruption was particularly marked. Almost two-thirds of participants received homebound or homeschool instruction, nearly 90% missed school because of illness, and most reported limitations in educational, social, and recreational activities. Because adolescence is a critical period for academic and psychosocial development, these disruptions may have lasting consequences. Previous studies have similarly identified ME/CFS as a major contributor to prolonged school absence and impaired educational participation [6,7]. These findings underscore the importance of timely recognition, appropriate school accommodations, and multidisciplinary support.

Strengths of this study include its multisite design, use of standardized validated instruments across multiple illness domains, and comparison with a large adult MCAM cohort [10–20]. Reporting effect sizes alongside statistical significance also aided interpretation of the magnitude of between-group differences.

Several limitations should be acknowledged. The adolescent cohort was relatively small. Recruitment from specialty clinics may limit generalizability to community populations; specialty-clinic recruitment could overrepresent adolescents with more severe or persistent illness who seek subspecialty care but could also underrepresent severity because participants had access to specialized evaluation, management, and treatment. Because most participating clinics primarily specialized in adult ME/CFS, the pediatric cohort may not represent the full

spectrum of pediatric disease. The cross-sectional design precludes assessment of illness trajectory or causal relationships, and comparisons with adults should be interpreted cautiously because adults had substantially longer illness duration and may differ in referral patterns, cumulative symptom burden, and healthcare utilization. The substantial imbalance in cohort size should also be considered. Longitudinal studies are needed to clarify age-related differences in symptom persistence and functional trajectories [23].

These findings underscore the importance of recognizing pediatric ME/CFS as a multisystem illness associated with substantial functional limitations and educational disruption. Greater clinician awareness, timely diagnosis, appropriate symptom management, and coordinated school accommodations may help reduce diagnostic delay and support affected adolescents.

In summary, adolescents with ME/CFS evaluated at specialty clinics exhibited substantial multisystem symptom burden, impaired physical functioning, and profound educational disruption. Although adults reported greater burden across several physical symptom domains, many differences were small in magnitude, indicating substantial similarity in core clinical features across age groups. Future longitudinal studies are needed to better understand disease progression and inform targeted interventions [23,24].

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CRediT authorship contribution statement

Stephanie Sargent Weaver: Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing.

Joanna J. Regan: Conceptualization, Investigation, Methodology, Writing – review & editing.

Anindita Issa: Conceptualization, Writing – original draft, Writing – review & editing.

Elizabeth A. Fall: Data curation, Resources, Writing – review & editing.

Yang Chen: Data curation, Resources, Writing – review & editing.

Jin-Mann S. Lin: Funding acquisition, Project administration, Resources, Validation, Writing – review & editing.

Elizabeth R. Unger: Funding acquisition, Project administration, Resources, Validation, Writing – review & editing.

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Table 1: Data Sources and Standardized Measures Used to Assess Clinical and Functional Characteristics of ME/CFS

Data Source	*Measure*
Sociodemographic characteristics	Age at enrollment, demographic questions
General clinical characteristics	Illness abstraction form (age at ME/CFS diagnosis, age at fatigue onset, age at chronic fatigue onset)
	Questionnaire items (# of visits, homeschooled, miss work or school, days missed in one month)
	CDC-Symptom Inventory (CDC-SI; questions #1-5, C12, C12a, C13, mode of onset) [11]
	Physical exam (to calculate BMI)
Functional impairment	Medical Outcomes Study Short Form 36 (SF-36v2) [13, 14]
Fatigue and post-exertional Malaise	CDC-SI (PEM, fatigue duration)
	20-item Multidimensional Fatigue Inventory (MFI-20) [15, 16]
Neurocognitive/autonomic symptoms	CDC-SI (forgetfulness, difficulty thinking or concentrating)
	DePaul Questionnaire (ringing in ears, problems remembering things, difficulty paying attention, difficulty finding the right word, difficulty understanding things, only able to focus on one thing at a time, frequently losing your train of thought, slowness of thought, absent-mindedness, recent trouble with math or numbers, feel unsteady on your feet, shortness of breath, dizziness, irregular heartbeats) [17-20]
Pain	CDC-SI (muscle aches or pains, joint pain, headaches)
Immunologic/inflammation symptoms	CDC-SI (sore throat, tender lymph nodes or swollen glands)

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Gastrointestinal	CDC-SI (stomach or abdominal pains)
	DePaul (upset stomach)
Emotional or behavioral	DePaul (mood changes, anxiety)
Sleep	CDC-SI (unrefreshing sleep)
Other	DePaul (some smells, foods or chemicals make you feel sick)

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Table 2: Sociodemographic Characteristics of Study Population

Sociodemographic Characteristics of Patients	N	ME/CFS 10-17 years
Age at enrollment mean (SD)	40	15.35 (1.89)
Sex n(%)	40	Females 21 (52.5%) Males 19 (47.5%)
Race n(%)	37	Other 7 (19%) White 30 (81%)
Ethnicity n(%)	32	Hispanic 1 (3.1%) Non-Hispanic 31 (96.9%)
Education (Grade in school) mean (SD)	39	9.53 (1.96)
Insurance status n(%)	36	Yes 36 (100%)
Parent/Guardian Marital Status		
Guardian 1	37	Married 30 (81.1%) Never married 1 (2.7%) Prev. married 6 (16.2%)
Guardian 2	28	Married 25 (89.3%) Prev. married 3 (10.7%)
Parent/Guardian Employment		
Guardian 1	33	Full-time 24 (72.7%) Not employed 7 (21.2%) Part-time 2 (6.1%)
Guardian 2	27	Full-time 11 (40.7%) Not employed 9 (33.3%) Part-time 7 (26%)
Parent or Guardian receive unemployment benefits	36	No 36 (100%)

Note: Percentages are based on non-missing responses. Sample sizes vary because of missing data.

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Table 3: General Clinical Characteristics of Patients

General Clinical Characteristics of patients	N	ME/CFS 10-17 years
Age at ME/CFS diagnosis, mean (SD)	24	13.71 (2.77)
Age at fatigue onset , **mean (SD)**	26	11.73 (2.52)
Age at chronic fatigue onset (≥6 months), **mean (SD)**	12	13.50 (1.93)
Mode of onset, n(%)	36	
Sudden		20 (55.6%)
Gradual		16 (44.4%)
BMI, mean (SD)	21	21.94 (4.59)
Clinic visits in the past 12 months, mean (SD)	36	2.06 (2.51), Range 0-9
Attend school or have homebound instruction, n(%)	39	School 14 (36%), Homebound/Homeschool 25 (64%)
Missed school due to illness, n(%)	36	Yes 32 (89%), No 4 (11%)
School days missed in the past 30 days, mean (SD)	29	10.78 (8.30)
Rest makes fatigue better, n(%)	40	Yes, a lot 4 (10%) Yes, a little 11 (27.5%) No, not very much 16 (40%) No, not at all 9 (22.5%)
Illness limited ability to pursue educational activities, n(%)	39	Yes 33 (85%), No 6 (15%)
Illness limited social activities, n(%)	40	Yes 32 (80%), No 8 (20%)

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Illness limited recreational activities (i.e., sports), n(%)	39	Yes 36 (92.3%), No 3 (7.7%)
Most bothersome symptom, n(%)	39	Fatigue, Tiredness or Exhaustion 17 (43.6%)
Other symptoms been part of ill health, n(%)	37	Yes 22 (59.5%), No 15 (40.5%)
Other symptoms, top three, n(%)	22	1. Dizziness/POTS symptoms 14 2.Anxiety/Depression/OCD 10 3. Muscles aches & pains 8

Note: Percentages are based on non-missing responses. Sample sizes vary because of missing data. Age at ME/CFS diagnosis and age at fatigue onset were missing or not reported for some participants.

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Table 4: Measures of Functional Impairment in Patients

Medical Outcomes Study Short Form 36 (SF-36)	**ME/CFS 10-17 yrs Mean (SD)**	**ME/CFS 18+ yrs Mean (SD)**	**General population reference Mean (SD)**	***P*-value**	***Hedges' g***
Physical component summary	30.6 (12.4)	25.6 (10.1)	50 (10)	0.0117*	0.50
Mental component summary	41.5 (11)	41.7 (12.7)	50 (10)	0.3341	-0.02
Physical functioning	35.6 (13.3)	31.2 (10.2)	50 (10)	0.0217*	0.42
Role physical	27.2 (11.6)	24.2 (8.6)	50 (10)	0.4495	0.35
Bodily pain	39 (12.7)	35.8 (10.3)	50 (10)	0.1863	0.30
Role vitality/energy/fatigue	36 (11.8)	30.3 (8.6)	50 (10)	0.0009***	0.64
General health	31.1 (10.9)	28.9 (8.5)	50 (10)	0.0628	0.25
Role emotional	40.3 (15.2)	42.6 (15.9)	50 (10)	0.3788	-0.15
Social functioning	29.4 (13.4)	24.4 (10.4)	50 (10)	0.0015**	0.48
Mental health	43.8 (12)	44.1 (12)	50 (10)	0.4535	-0.02

*p<0.05, **p<0.01, ***p<0.001

Note: SF-36v2 scores are reported as norm-based T-scores, standardized to a general-population mean of 50 and standard deviation of 10; higher scores indicate better health and functioning. Effect sizes were calculated using Hedges' g to account for unequal group sizes and the small pediatric sample.

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Table 5: Measures of Fatigue and Post-Exertional Malaise (PEM) in Patients

Measure	ME/CFS 10-17 yrs Mean (SD)	ME/CFS 18+ yrs Mean (SD)	*P*-value	Hedges' *g*
CDC-SI (score range 0-16)				
Fatigue duration, years	4.00 (3.42)	13.62 (10.25)	< .0001***	-0.96
Unusual fatigue for at least one day after exertion (PEM)	7.94 (6.61)	11.28 (5.15)	0.0010***	-0.64
PEM present ≥6 months, n (%)	26 (66.7%)	515 (87.7%)	0.0002**	--
Multidimensional Fatigue Inventory (MFI-20)				
General fatigue	16.53 (3.86)	18.15 (2.39)	0.0030**	-0.65
Physical fatigue	15.74 (4.07)	17.34 (3.03)	0.0099**	-0.51
Reduced activity	14.40 (4.97)	16.12 (3.71)	0.0283*	-0.45
Reduced motivation	12.18 (3.86)	12.10 (4.24)	0.4627	0.02
Mental fatigue	14.55 (3.70)	14.64 (3.98)	0.3572	-0.02

*p<0.05, **p<0.01, ***p<0.001

Note: Symptom presence for CDC Symptom Inventory (CDC-SI) variables was defined as symptoms reported during the preceding month that had persisted for at least 6 months. Mean CDC-SI scores reflect symptom frequency and severity during the preceding month. MFI-20 scores range from 4 to 20, with higher scores indicating a higher level of fatigue. Effect sizes were calculated using Hedges' *g* to account for unequal group sizes and the small pediatric sample.

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Table 6: Measures of Neurocognitive and Autonomic Symptoms in Patients

Presence ≥6 months, n (%)				Mean score (SD)			
Symptom	**ME/CFS 10-17 yrs**	**ME/CFS 18+ yrs**	***p*-value**	**ME/CFS 10-17 yrs**	**ME/CFS 18+ yrs**	***P*-value**	**Hedge's g**
CDC-SI (score range 0-16)							
Forgetfulness or memory problems	22 (55.0%)	373 (63.8%)	0.2663	6.56 (7.01)	6.39 (5.72)	0.1309	-0.18
Difficulty thinking or concentrating	27 (67.5%)	422 (72.1%)	0.5282	7.37 (7.33)	7.52 (5.61)	0.4912	-0.01
DePaul (score range 0-16)							
Problems remembering things	27 (67.5%)	503 (84.5%)	0.0050**	8.74 (6.02)	6.87 (5.08)	0.047*	0.36
Difficulty paying attention	28 (70%)	504(84.7%)	0.0146*	9.76 (6.05)	7.55 (5.26)	0.0192*	0.42
Difficulty finding the right word	19 (47.5%)	500(84%)	<0.0001***	4.77 (5.96)	6.47 (5.14)	0.0020**	-0.33
Difficulty understanding	20 (50%)	438(73.6%)	0.0013**	5.14 (5.66)	4.04 (4.38)	0.4940	0.25
Only focus on one thing	20 (50%)	488(82%)	<0.0001***	5.74 (6.34)	6.91 (5.25)	0.0283*	-0.22
Lose train of thought	24 (60%)	—	—	6.15 (5.37)	—	—	—
Slowness of thought	21 (52.5%)	479(80.5%)	<0.0001***	7.06 (6.75)	5.62 (5.11)	0.3762	0.27
Absentmindedness or forgetfulness	25 (62.5%)	494(83%)	0.0011**	7.33 (5.99)	6.21 (5.26)	0.2418	0.21
Recent trouble with math or numbers	17 (42.5%)	—	—	6.61 (7.35)	—	—	—
Unsteady on feet	14 (35%)	452(76%)	<0.0001***	4.11 (6.19)	3.23 (3.98)	0.0251*	0.21
Shortness of breath	17 (42.5%)	387(65%)	0.004**	3.94 (4.81)	2.94 (3.97)	0.4857	0.25

Dizziness	20 (50%)	420(70.6 %)	0.006**	5.21 (6.03)	2.74 (3.41)	0.1320	0.69
Irregular heartbeats	10 (25%)	319(53.6 %)	0.0005**	2.94 (5.57)	1.71 (2.85)	0.0977	0.40
Ringing in ears	30 (75%)	—	—	7.94 (6.61)	—	—	—

—Not assessed in adult study population *p<0.05, **p<0.01, ***p<0.001

Note: For CDC Symptom Inventory (CDC-SI) variables, symptom presence was defined as symptoms reported during the preceding month that had persisted for at least 6 months. Symptom presence for DePaul Symptom Questionnaire (DSQ) variables was based on the harmonized 6-month symptom measures. Mean CDC-SI scores reflect symptom frequency and severity during the preceding month. Effect sizes were calculated using Hedges' *g* to account for unequal group sizes and the small pediatric sample.

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Table 7: Measures of Pain Symptoms in Patients

Presence ≥6 months, n (%)				Mean score (SD)			
Symptom	**ME/CFS 10-17 yrs**	**ME/CFS 18+ yrs**	***P*-value**	**ME/CFS 10-17 yrs**	**ME/CFS 18+ yrs**	***P*-value**	**Hedges' g**
CDC-SI (score range 0-16)							
Muscle aches or pain	23 (57.5%)	469 (79.9%)	0.0009***	5.12 (5.57)	8.28 (5.50)	0.0004***	-0.57
Joint pain	20 (51.3%)	364 (62.0%)	0.1828	4.45 (5.73)	5.89 (5.63)	0.0411*	-0.26
Headaches	21 (52.5%)	377 (64.3%)	0.1324	4.94 (5.96)	4.96 (4.92)	0.2022	-0.01

*p<0.05, **p<0.01, ***p<0.001

Note: Symptom presence for CDC Symptom Inventory (CDC-SI) variables was defined as symptoms reported during the preceding month that had persisted for at least 6 months. Mean CDC-SI scores reflect symptom frequency and severity during the preceding month. Effect sizes were calculated using Hedges' *g* to account for unequal group sizes and the small pediatric sample.

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Table 8: Measures of Other Symptoms in Patients

Presence ≥6 months, n (%)				Mean score (SD)			
Symptom	**ME/CFS 10-17 yrs**	**ME/CFS 18+ yrs**	***P*-value**	**ME/CFS 10-17 yrs**	**ME/CFS 18+ yrs**	***P*-value**	**Hedges' g**
Immunologic/inflammatory							
CDC-SI (score range 0-16)							
Sore throat	5 (12.5%)	202 (34.4%)	0.0044**	0.84 (2.50)	1.97 (3.32)	0.0016**	-0.35
Tender Lymph nodes or swollen glands in your neck or armpits	8 (20.5%)	288 (49.2%)	0.0005**	1.60 (3.66)	3.58 (4.66)	0.0003***	-0.43
Gastrointestinal							
CDC-SI (score range 0-16)							
Stomach or abdominal pains	19 (47.5%)	241 (41.2%)	0.4339	3.05 (4.27)	2.95 (4.37)	0.2711	0.02
DePaul (score range 0-16)							
Upset stomach	24(60%)	—	—	6.88 (5.73)	—	—	—
Emotional or behavioral							
DePaul (score range 0-16)							
Mood changes	20(50%)	—	—	5.44 (5.47)	—	—	—
Anxiety	23(57.5%)	—	—	7.55 (6.49)	—	—	—
Sleep							
CDC-SI (score range 0-16)							
Unrefreshing sleep	29 (72.5%)	509 (87.0%)	0.0103*	8.98 (6.81)	11.08 (5.31)	0.040*	-0.39
Other							
DePaul (score range 0-16)							
Some smells, foods or chemicals make you feel sick	29(72.5%)	408(68.6%)	0.6036	5.12 (5.57)	4.77 (5.33)	0.3505	0.14

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—Not assessed in adult study population *p<0.05, **p<0.01, ***p<0.001
Note: For CDC Symptom Inventory (CDC-SI) variables, symptom presence was defined as symptoms reported during the preceding month that had persisted for at least 6 months. Symptom presence for DePaul Symptom Questionnaire (DSQ) variables was based on the harmonized 6-month symptom measures. Mean CDC-SI scores reflect symptom frequency and severity during the preceding month. Effect sizes were calculated using Hedges' *g* to account for unequal group sizes and the small pediatric sample.

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Declaration of Interest

Manuscript Title:
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Authors:
Stephanie Sargent Weaver, Joanna J. Regan, Anindita Issa, Elizabeth A. Fall, Yang Chen, Jin-Mann S. Lin, Elizabeth R. Unger

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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